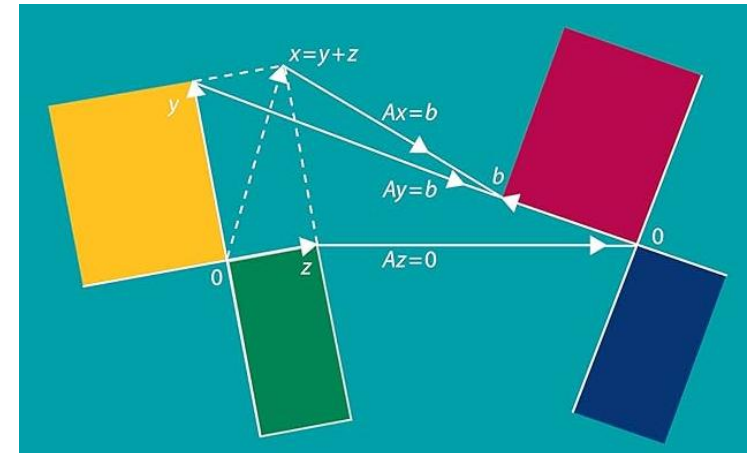


Vector Spaces

(Linear Algebra)

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Basis and Spanning Set

To describe a vector in $\mathbb{R}^n$ we need a coordinate system. A basis is a coordinate system or framework which describes the Euclidean n -space.

For example, there are infinitely many vectors in the plane $\mathbb{R}^2$, but we can describe all of these by using the standard unit vectors $\mathbf{e}_1 = (1 \ 0)^T$ in the x direction and $\mathbf{e}_2 = (0 \ 1)^T$ in the y direction.

We can write a vector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$, but what does 2 and 3 represent?

It indicates two units in the x direction and three units in the y direction which can be written as $2\mathbf{e}_1 + 3\mathbf{e}_2$.

Let $\mathbf{v} = (a \ b \ c)^T$ be any vector in $\mathbb{R}^3$. Write this vector $\mathbf{v}$ in terms of the unit vectors:

$$\mathbf{e}_1 = (1 \ 0 \ 0)^T, \ \mathbf{e}_2 = (0 \ 1 \ 0)^T \text{ and } \mathbf{e}_3 = (0 \ 0 \ 1)^T$$

Vectors $\mathbf{e}_1, \mathbf{e}_2$ and $\mathbf{e}_3$ specify x, y and z directions respectively. (Illustrated in Fig. 2.26.)

Solution

We have

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3$$

That is, we can write the vector $\mathbf{v}$ as a linear combination of vectors $\mathbf{e}_1, \mathbf{e}_2$ and $\mathbf{e}_3$.

Definition (2.25). Consider the n vectors in the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ in the n -space, $\mathbb{R}^n$. If every vector in $\mathbb{R}^n$ can be produced by a linear combination of these vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ then we say these vectors **span** or **generate** the n -space, $\mathbb{R}^n$.

This set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ is called the **spanning set**. We also say that the set S spans the n -space or S spans $\mathbb{R}^n$.

For example, the standard unit vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ span $\mathbb{R}^2$ because $k\mathbf{e}_1$ spans the x axis and $c\mathbf{e}_2$ spans the y axis. Hence, by introducing scalars, k and c , the linear combination, $k\mathbf{e}_1 + c\mathbf{e}_2$, of these vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ can produce any vector in $\mathbb{R}^2$.

Basis

Given some vectors we can generate others by a linear combination. We need *just* enough vectors to build all other vectors from them through linear combination. This set of just enough vectors is called a basis.

An example is the standard unit vectors $\mathbf{e}_1 = (1 \ 0)^T$, $\mathbf{e}_2 = (0 \ 1)^T$ for $\mathbb{R}^2$. This is the basis which forms the x and y axes of $\mathbb{R}^2$ because $\mathbf{e}_1 = (1 \ 0)^T$ specifies the x direction and $\mathbf{e}_2 = (0 \ 1)^T$ specifies the y direction.

Each additional basis vector introduces a new direction.

Definition (2.26). Consider the n vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ in the n space, $\mathbb{R}^n$.

These vectors form a **basis** for $\mathbb{R}^n \Leftrightarrow$

- (i) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ span $\mathbb{R}^n$ and
- (ii) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ are linearly independent

Basis

We can write the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ as a set $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$. These are called the **basis vectors** – independent vectors which span $\mathbb{R}^n$. Any vector in $\mathbb{R}^n$ can be constructed from the basis vectors.

Bases (plural of basis) are the most efficient spanning sets. There are many sets of vectors that can span a space. However, in these sets some of the vectors might be redundant in spanning the space (because they can be ‘made’ from the other vectors in the set). A basis has no redundant vectors. This is exactly what is captured by demanding linear independence in the definition.

Spanning Set VS Basis Set

- A spanning set is any collection of vectors that can generate the whole vector space.
- A basis is the smallest such collection with no unnecessary vectors.

Spanning Set	Basis Set
Can generate (span) the entire vector space.	
May contain redundant vectors	Contains no redundant vectors.
Vectors may be linearly dependent.	Vectors must be linearly independent.
Every basis is a spanning set.	Not every spanning set is a basis.

Vector Spaces

- Let V be a non-empty set of elements called vectors. We define two operations on the set V — vector addition and scalar multiplication. Scalars are real numbers.
- Let u , v and w be vectors in the set V . The set V is called a vector space if it satisfies the following 10 axioms.
 1. The vector addition $u + v$ is also in the vector space V . Generally in mathematics we say that we have closure under vector addition if this property holds.
 2. Commutative law: $u + v = v + u$.

Vector Spaces

- 3. **Associative law:** $(u + v) + w = u + (v + w)$.
- 4. **Neutral element.** There is a vector called the zero vector in V denoted by O which satisfies

$$u + O = u \text{ for every vector } u \text{ in } V$$

- 5. **Additive inverse.** For every vector u there is a vector $-u$ (minus u) which satisfies the following:

$$u + (-u) = O$$

Vector Spaces

6. Let k be a real scalar then ku is also in V .

7. Associative law for scalar multiplication. Let k and c be real scalars then

$$k(c\mathbf{u}) = (kc)\mathbf{u}$$

8. Distributive law for vectors. Let k be a real scalar then

$$k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$$

9. Distributive law for scalars. Let k and c be real scalars then

$$(k + c)\mathbf{u} = k\mathbf{u} + c\mathbf{u}$$

10. Identity element. For every vector $\mathbf{u}$ in V we have

$$1\mathbf{u} = \mathbf{u}$$

Vector Spaces (Simplified)

- A vector space is a collection of vectors where vector addition and scalar multiplication always produce another vector in the same collection.
- Example:
 - Imagine a football field. Players can run anywhere and pass the ball. They can move around. So everything stays inside the field. A vector space is similar.
 - If we leave the field then It is not a vector space.

Vector Subspaces

Let V be a vector space and S be a non-empty subset of V . If the set S satisfies all 10 axioms of a vector space with respect to the same vector addition and scalar multiplication as V then S is also a vector space. We say S is a subspace of V .

Definition (3.4). A non-empty subset S of a vector space V is called a **subspace** of V if it is also a vector space with respect to the same vector addition and scalar multiplication as V .

We illustrate this in Fig. 3.3.

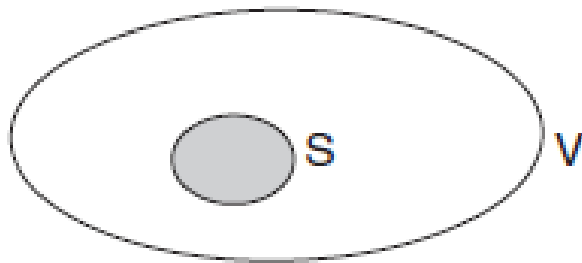


Figure 3.3

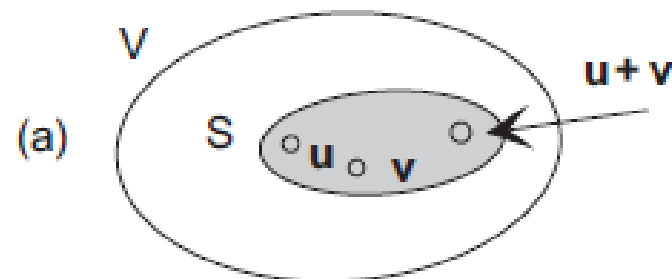
Note the difference between subspace and subset. A subset is merely a specific set of elements chosen from V . A subset must also satisfy the 10 axioms of vector space to be called a **subspace**.

Vector Subspaces

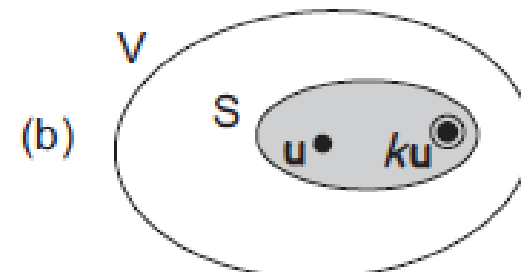
Proposition (3.5). Let S be a non-empty subset of a vector space V . Then S is subspace of $V \Leftrightarrow$:

- (a) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in the set S then the vector addition $\mathbf{u} + \mathbf{v}$ is also in S .
- (b) If $\mathbf{u}$ is a vector in S then for every scalar k we have, $k\mathbf{u}$ is also in S .

Note that this proposition means that we must have *closure* under both vector addition and scalar multiplication. This means that S is a subspace of $V \Leftrightarrow$ both the following are satisfied, as shown in Fig. 3.5.



Vector addition



Scalar multiplication

Vector Subspace (Simplified)

- A vector subspace is a smaller vector space inside a larger vector space.
- Example:
 - The larger vector space is a big playground.
 - A subspace is a smaller playground inside it.
 - You can still perform all vector operations (addition and scalar multiplication) without leaving the smaller playground.

Column Space

- The column space of a matrix is the set of all possible vectors that can be created by combining the columns of the matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$c_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

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$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Null Space

- The null space is the set of all vectors that become the zero vector when multiplied by the matrix.
- The null space contains all solutions to the equation $A\mathbf{x} = \mathbf{0}$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}.$$

Orthogonality and Orthogonal Vectors

- Orthogonal = Perpendicular
- Two vectors are orthogonal if the angle between them is 90° .

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Two vectors are orthogonal if their dot product is zero.

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Orthogonal

- The vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal if their dot product is equal to 0.

$$\begin{aligned}\vec{x} &= [x_1, x_2, \dots, x_n] \\ \vec{y} &= [y_1, y_2, \dots, y_n]\end{aligned}$$

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$$\vec{x} \cdot \vec{y} = 0$$

Orthonormal

- The word **orthonormal** is made up of two parts:
 - **Ortho** = **Orthogonal** (Perpendicular)
 - **Normal** = **Normalized** (Unit length)
- A set of vectors is orthonormal if every vector has length 1 and every pair of different vectors is perpendicular to each other.

Orthonormal Matrix

- Q is orthonormal if the following conditions are satisfied:

$$\begin{aligned} \|\vec{q}_i\|_2^2 &= 1 & \forall i \in [1, n] \\ \vec{q}_i \cdot \vec{q}_j &= 0 & \forall i, j \in [1, n] \text{ \& } i \neq j \end{aligned}$$

- The magnitude of every column of the orthonormal matrix is 1, and each column is perpendicular to the other.

$$\begin{aligned} Q &= \begin{bmatrix} q_{11} & q_{12} & \dots & q_{1n} \\ q_{21} & q_{22} & \dots & q_{2n} \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ q_{n1} & q_{n2} & \dots & q_{nn} \end{bmatrix} \\ &= [\vec{q}_1 \quad \vec{q}_2 \quad \dots \quad \vec{q}_n] \end{aligned}$$

Example 1

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

✓ Dot product

$$\mathbf{u} \cdot \mathbf{v} = 0$$

✓ Lengths

$$\|\mathbf{u}\| = 1, \quad \|\mathbf{v}\| = 1.$$